

Session Objectives

- Understand spinal nerve injuries (radiculopathies) and their relationships with myotomes and dermatomes
- Review the basic architecture of the brachial plexus, roots to branches
- Identify upper limb neurological deficiencies by specific muscle and sensory deficits

Things to Remember

- Brachial plexus, distribution of peripheral nerves, muscles attachments and actions
- This is not a complete survey of brachial plexus neuropathies or symptoms
- Variations in the brachial plexus may cause variation in signs and symptoms associated with injury
- A “nerve” is just a bundle of axons, just because a specific nerve is damaged doesn’t mean all the axons are non-functional

Median nerve:

C6, C7, C8, T1

Anterior compartment of forearm

all except: flexor carpi ulnaris
and ulnar two digitations of
flexor digitorum

Anterior compartment of hand

Muscles of the thenar eminence

Opponens pollicis

Abductor pollicis brevis

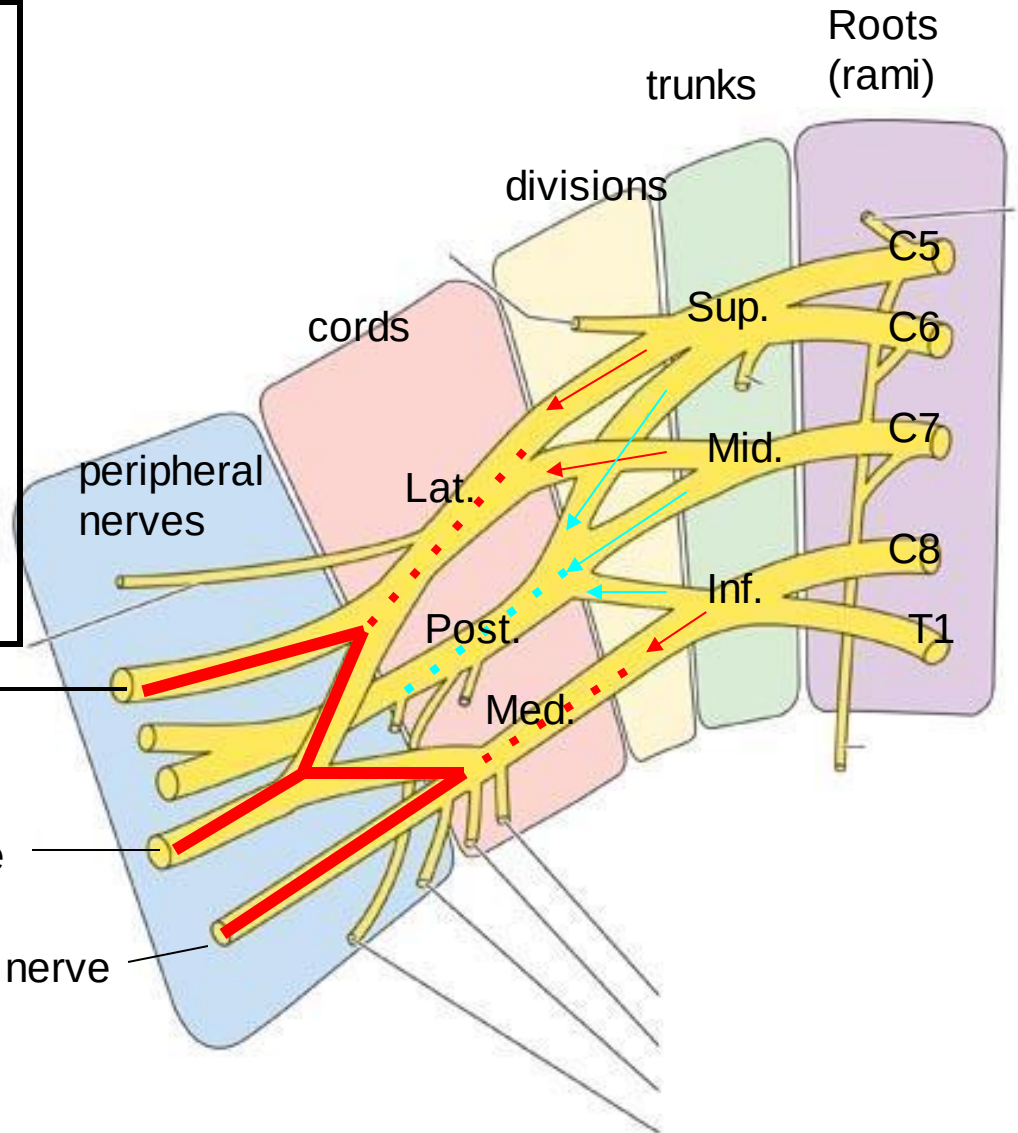
Flexor pollicis brevis

Radial two lumbricals

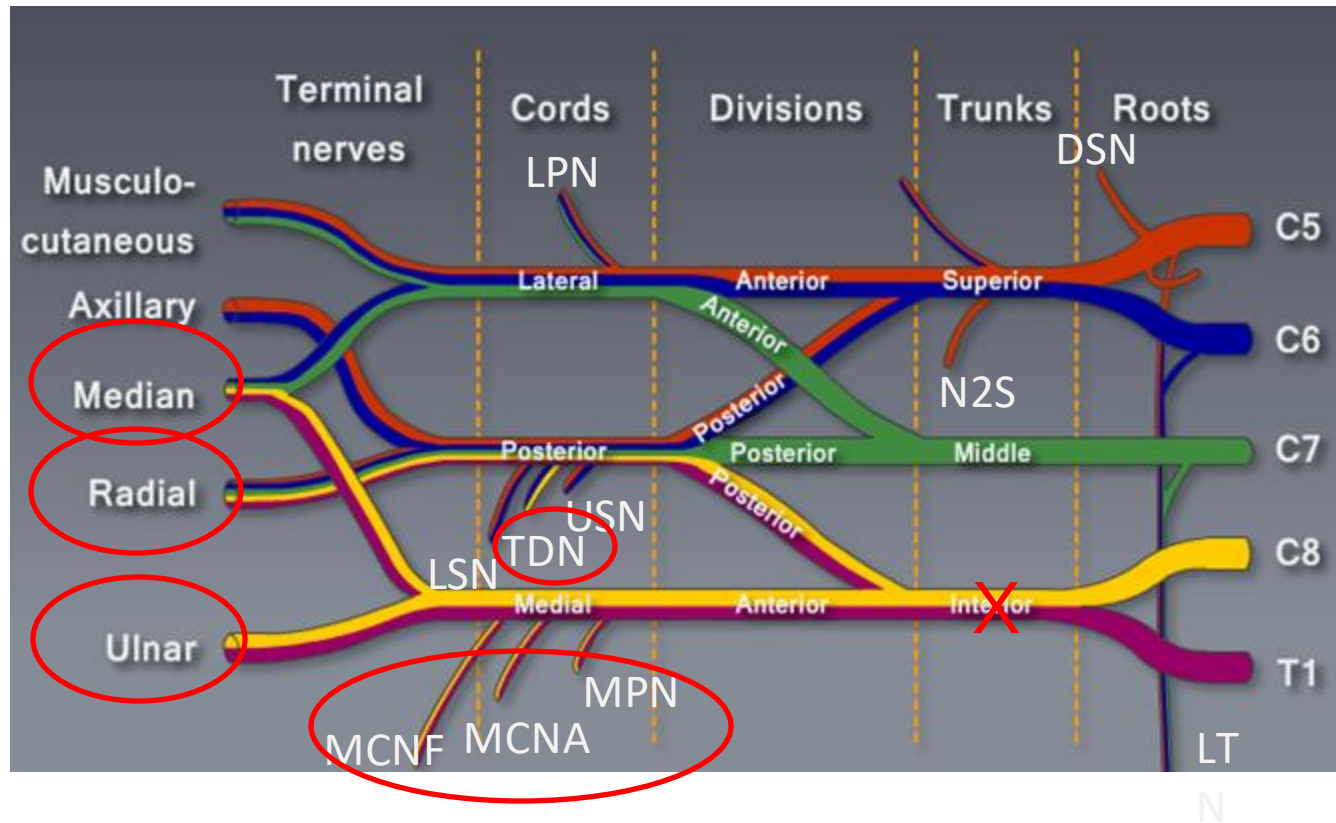
Musculocutaneous nerve

Median nerve

Ulnar nerve



Lesion of inferior trunk



Complete loss of function:

medial pectoral nerve
 medial cutaneous nerve of arm
 medial cutaneous nerve of forearm
 ulnar nerve

Partial loss of function:

thoracodorsal nerve
 median nerve
 radial nerve

Injuries to the superior trunk (C5-C6) of the brachial plexus (Erb-Duchenne Palsy):

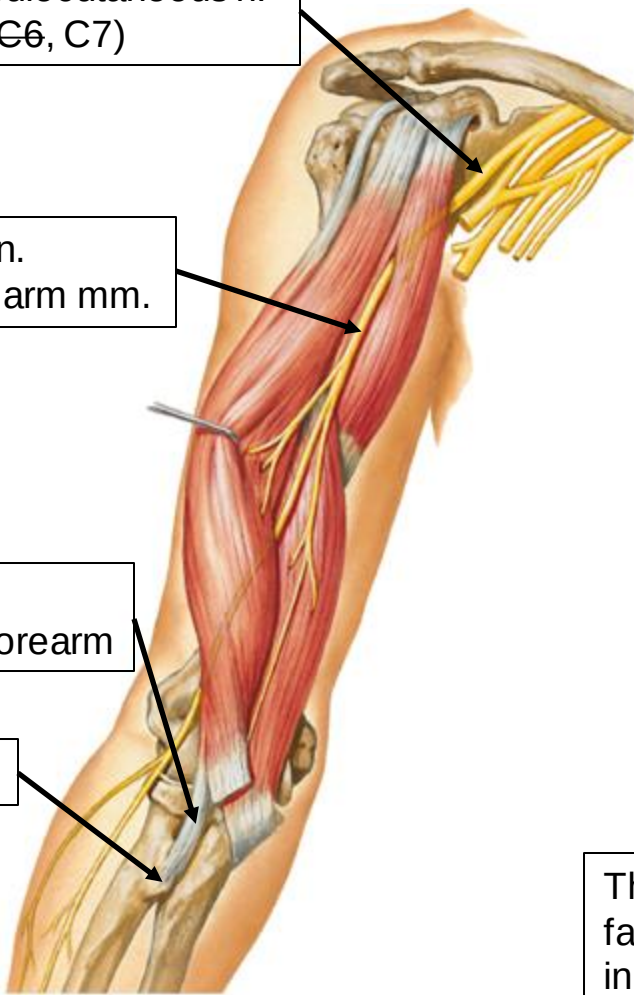
loss of supination of forearm from biceps brachii m.
due to damage to C5-C6 of musculocutaneous n.

Musculocutaneous n.
(C5, C6, C7)

Musculocutaneous n.
innervating anterior arm mm.

Biceps brachii m.:
major supinator of forearm

Radial tuberosity



Forearm in pronation due to
loss of major supinator of forearm

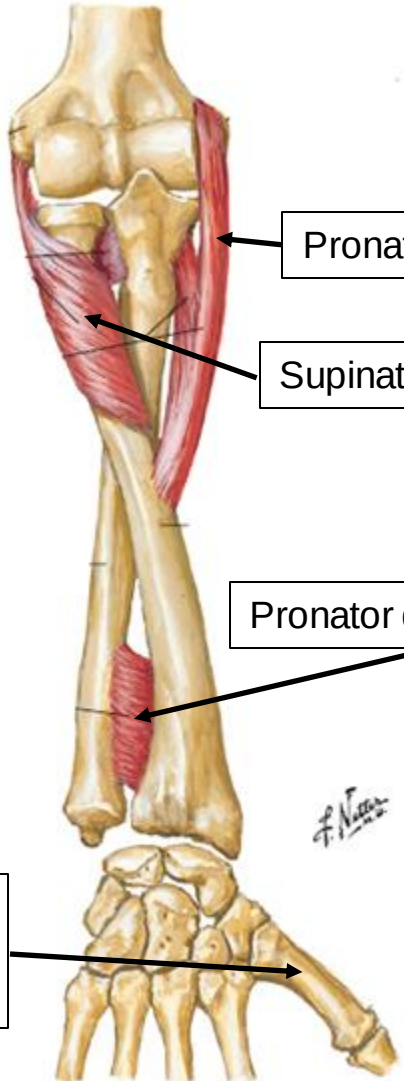
Pronated position

Pronator teres m.

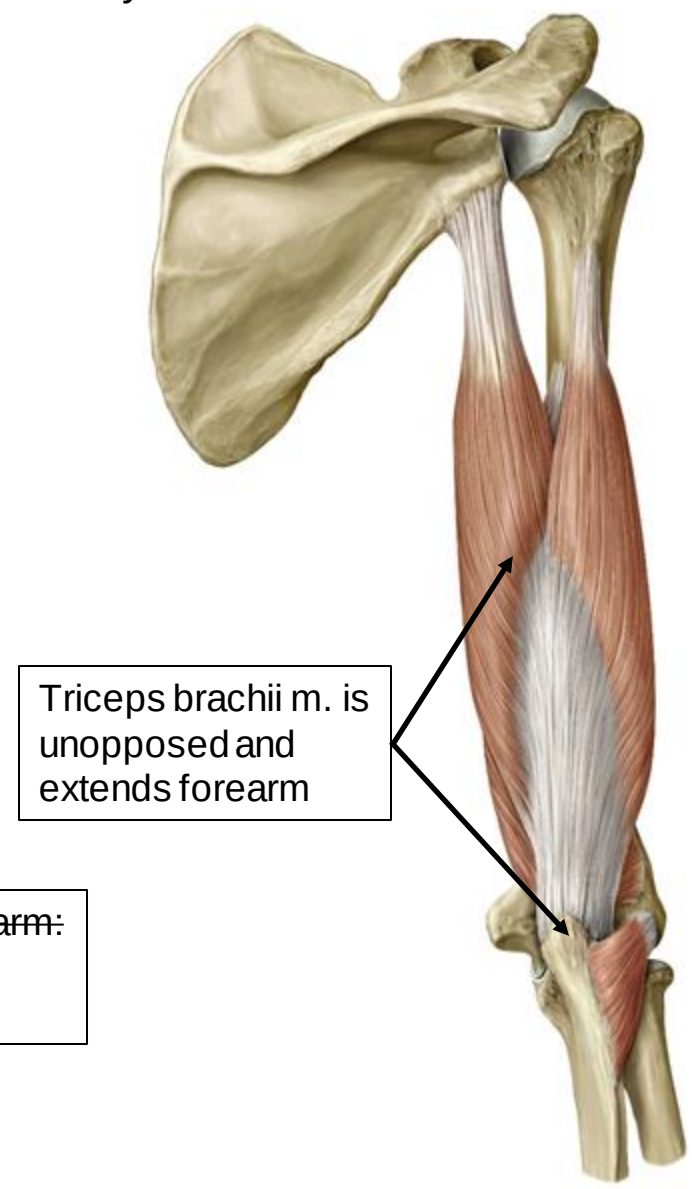
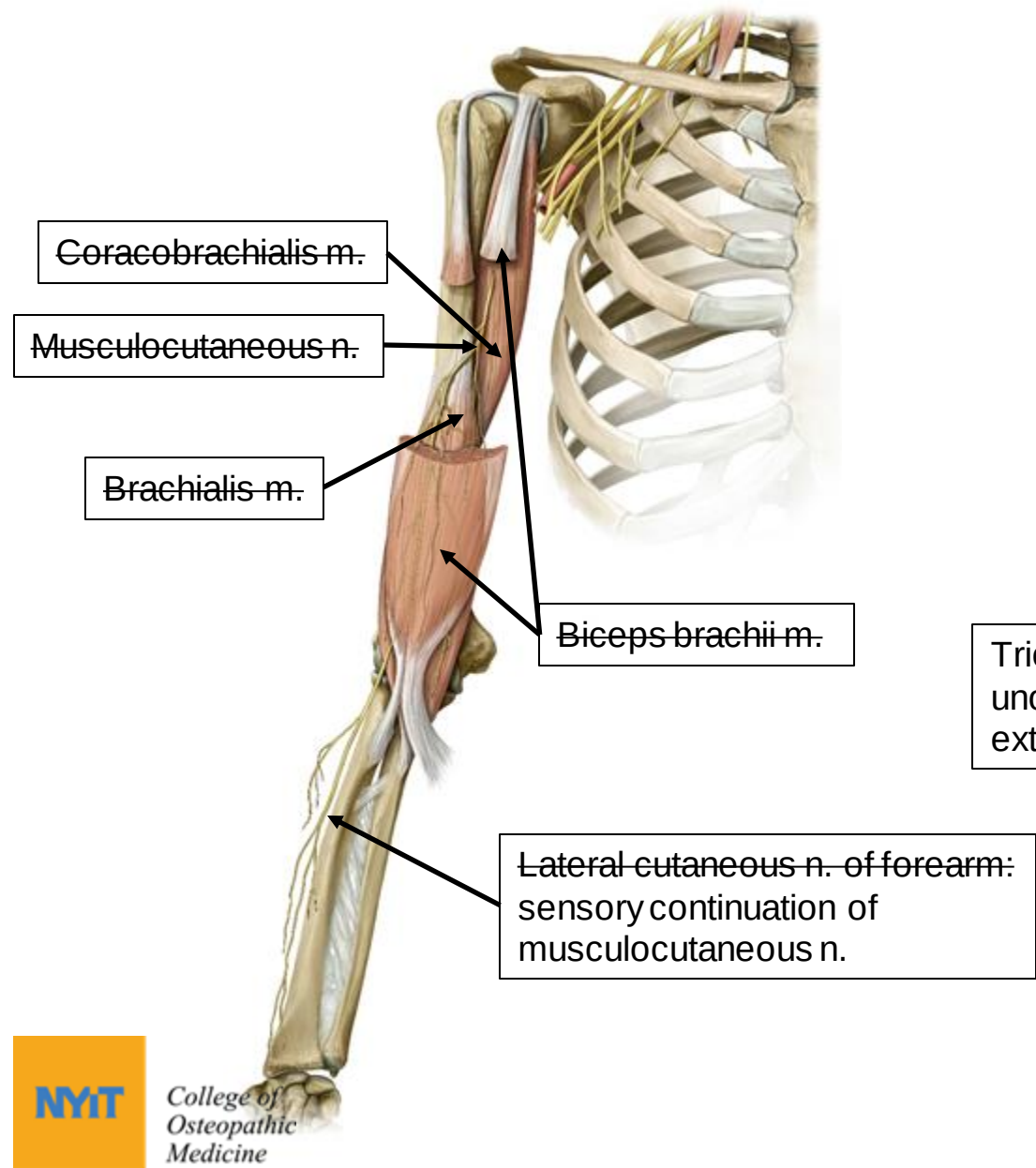
Supinator m.

Pronator quadratus m.

Thumb will be
facing medially
in pronation



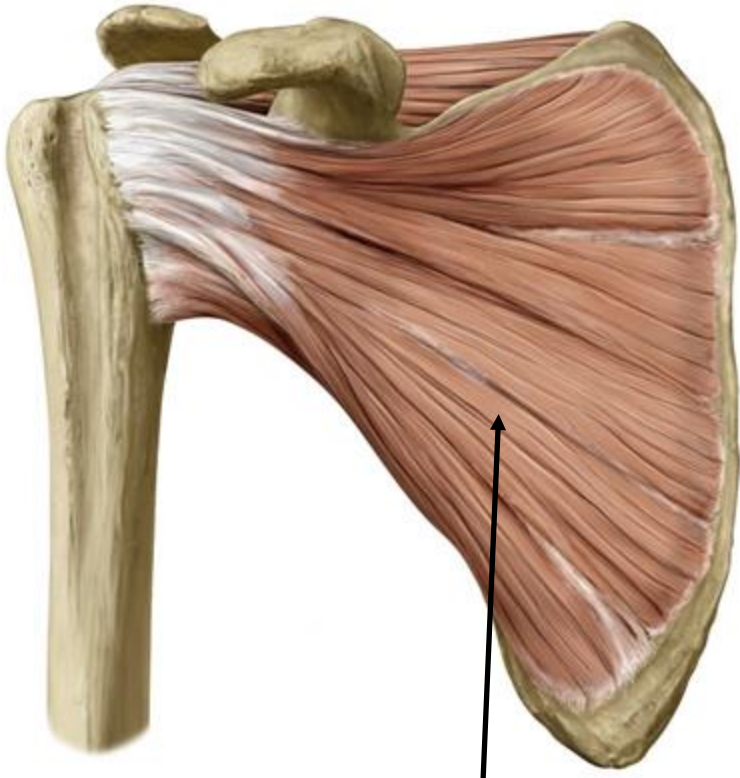
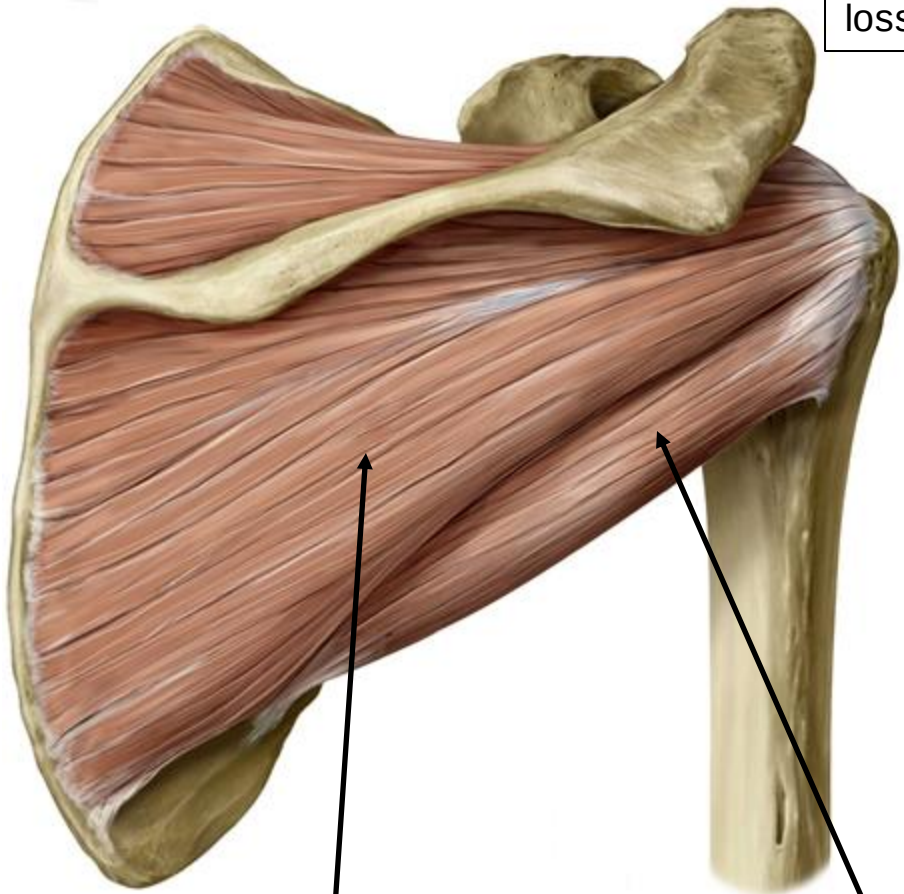
Injuries to the superior trunk (C5-C6) of the brachial plexus (Erb-Duchenne Palsy):
Causes loss of musculocutaneous n. function.
Loss of flexion of forearm from anterior arm mm. and sensory deficits to lateral side of forearm



Injuries to the superior trunk (C5-C6) of the brachial plexus (Erb-Duchenne Palsy):

Loss of lateral rotation of arm by infraspinatus m. and teres minor m. due to damage to C5-C6 of suprascapular n. and axillary n.

Arm is medially rotated due to loss of lateral rotators of arm



Infraspinatus m.:
1. Loss of innervation by suprascapular n. (C5-C6)
2. Loss of lateral rotation of arm

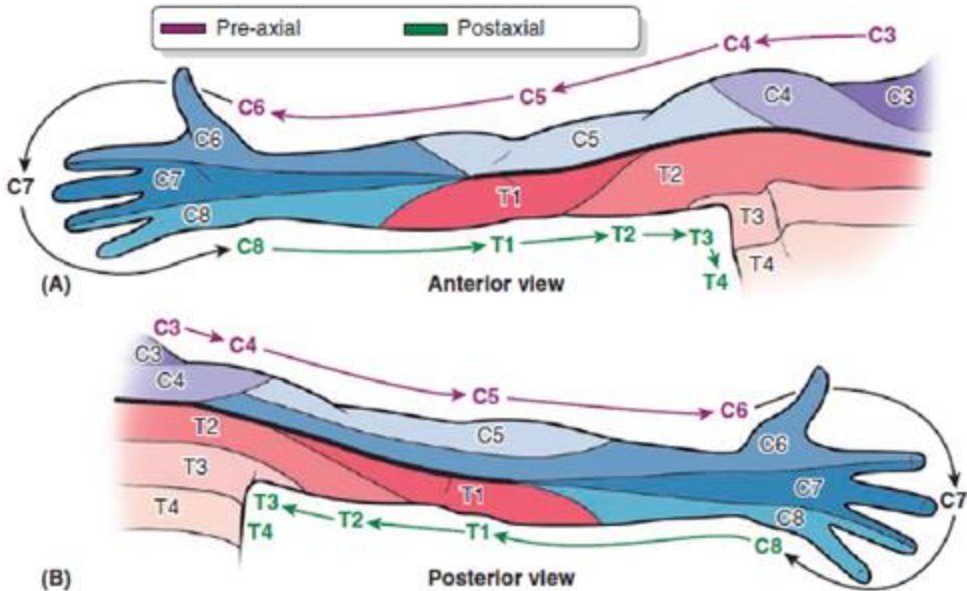
Teres minor m.:
1. Loss of innervation by axillary n. (C5-C6)
2. Loss of lateral rotation of arm

Subscapularis m.:
1. Innervated by subscapular nn. from posterior cord (C5-C6)
2. Arm still medially rotated from pectoralis major m.

Injuries to the superior trunk (C5-C6) of the brachial plexus (Erb-Duchenne Palsy):

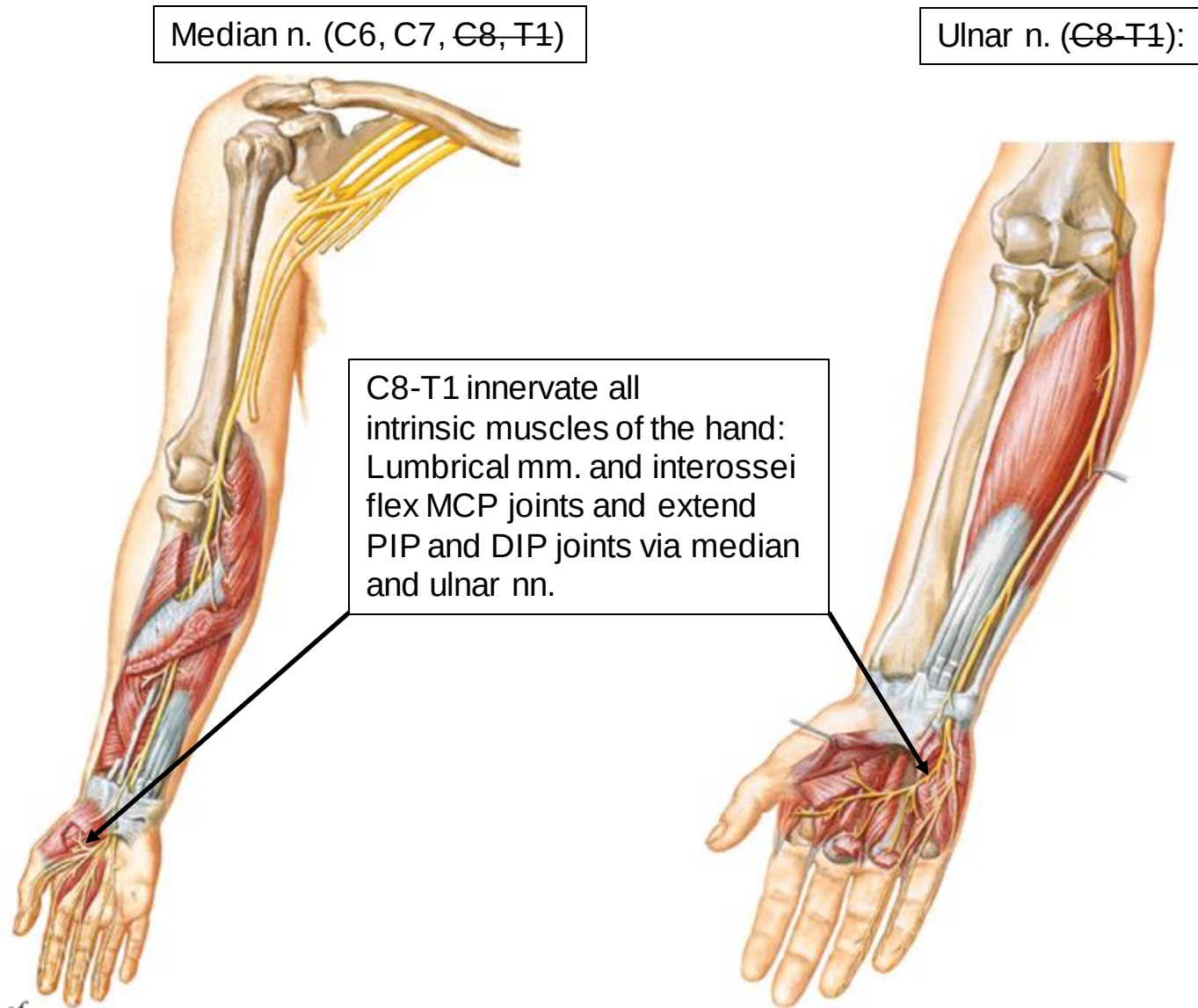
Clinical signs of damage to upper trunk:

- 1. Loss of lateral rotation of arm
- 2. Loss of flexion of forearm and supination impaired
- 3. Pronated forearm, medially rotated arm, and extended forearm due to unopposed action of pronators, pectoralis major m., and triceps brachii m., respectively, result in "waiter's tip position".
- 4. Sensory deficits on lateral aspect of shoulder, arm, forearm, and hand.



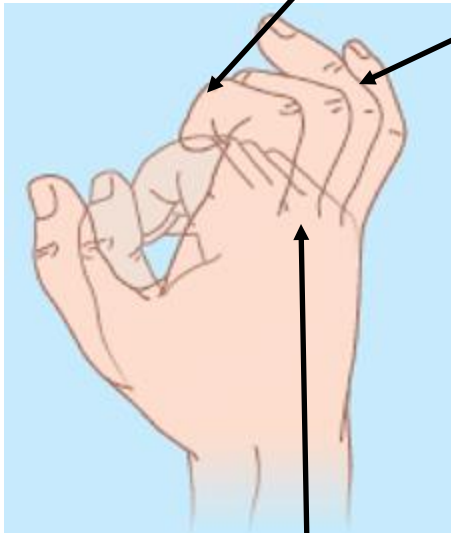
Damage to inferior trunk of brachial plexus (C8-T1): Klumpke's Palsy

Intrinsic hand muscles innervated by C8-T1



Damage to inferior trunk of brachial plexus (C8-T1): Klumpke's Palsy

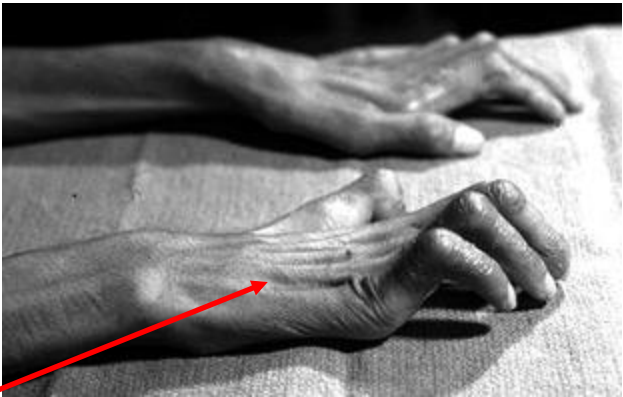
Results in "claw hand" due to loss of action of lumbrical mm. and interossei.



1. Flexion at PIP joints in digits 2-3 occurs due to unopposed action of flexor digitorum superficialis and due to loss of lumbrical and interossei mm. Flexion of DIP joints depends on functionality of lateral side of flexor digitorum profundus m. (C8)

2. PIP of digits 4-5 flexed but DIP not flexed because of unopposed action of FDS but lack of action from the medial side of FDP

Hyperextension of MCP joints occurs due to unopposed action of extensor digitorum m. due to loss of lumbrical mm.



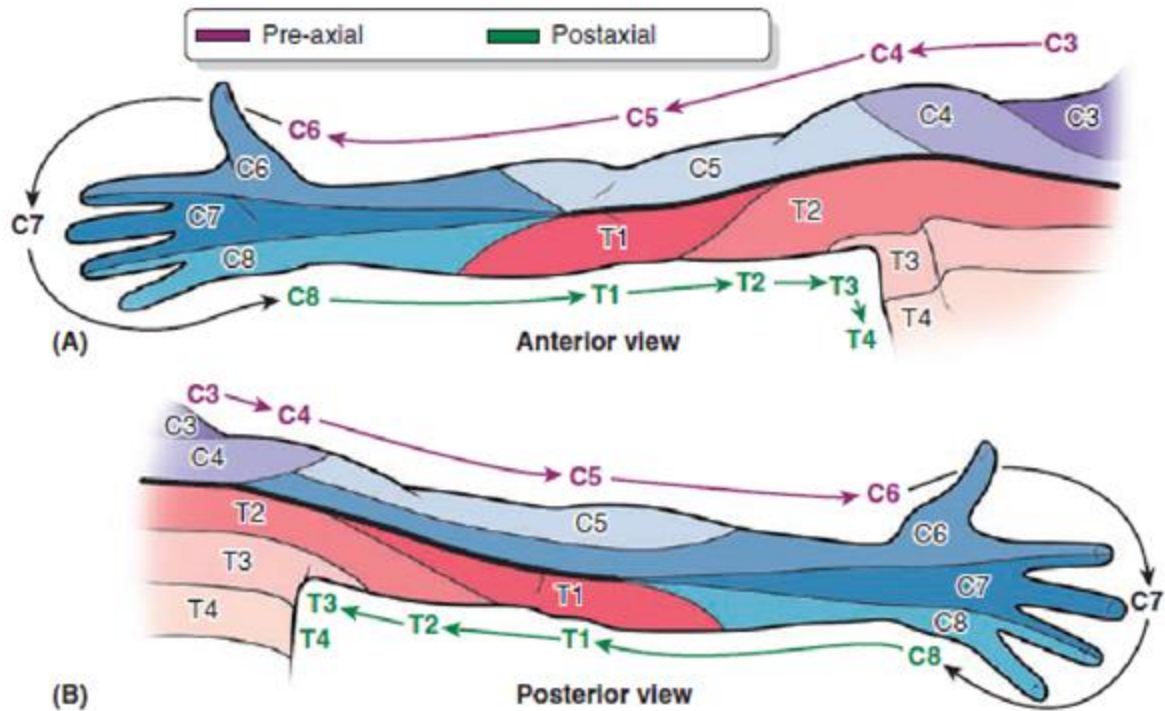
Over time, areas between metacarpals hollows out due to atrophy of lumbrical and interossei mm.

Schreuders et al. (2007), *Physikalische Medizin Rehabilitationsmedizin Kurortmedizin*, 17, 20-27, Fig. 1

Damage to inferior trunk of brachial plexus (C8-T1): Klumpke's Palsy

Clinical signs of Klumpke's Palsy:

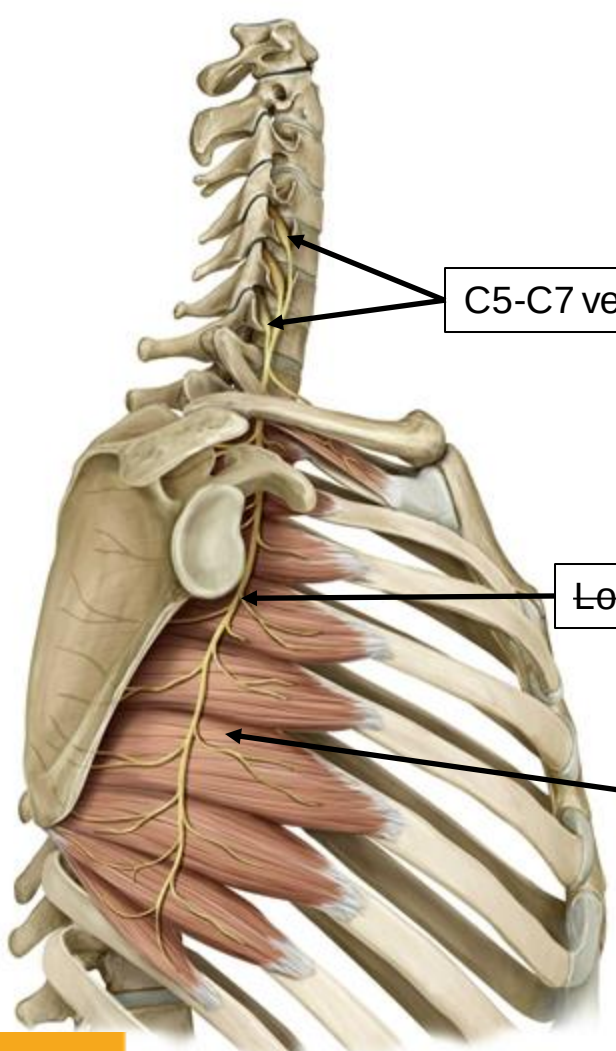
- 1. Loss of all lumbrical and interosseous mm. results in loss of abduction and adduction of fingers; hyperextension of MCP joints due to unopposed action of extensor digitorum m.
- 2. Loss of all lumbrical and interosseous mm. results in flexion of PIP joints in digits 2-5 due to unopposed action of intact flexor digitorum superficialis m. Flexion of DIP joints in digits 1-2 depends on functionality of the lateral side of flexor digitorum profundus mm. and severity of injury.
- 3. Total claw hand results from loss of all intrinsic hand muscles innervated by C8-T1 fibers from ulnar and median nn.
- 4. Radial deviation in wrist flexion from loss of action of flexor carpi ulnaris m.
- 5. Loss of cutaneous innervation of medial arm and medial forearm and medial hand



Damage to long thoracic n. (C5-C7):

Radical mastectomy or knife wound to lateral chest

- 1. Winged scapula (medial border of scapula protrudes posteriorly)
- 2. Inability to abduct arm above horizontal plane due to loss of rotation of scapula



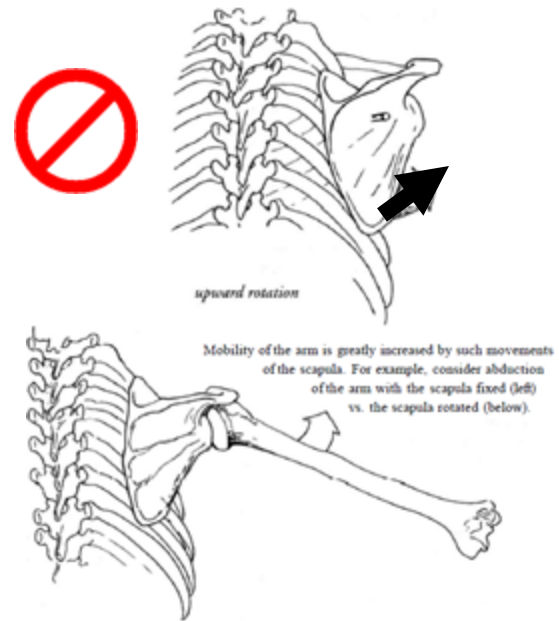
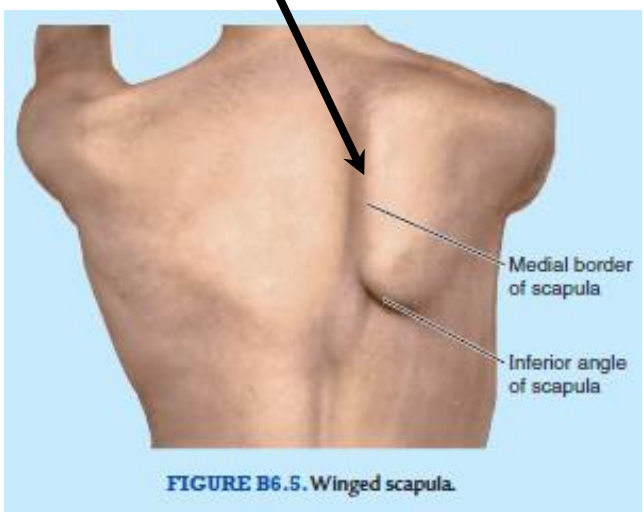
C5-C7 ventral rami

Long thoracic n.

Serratus anterior m.



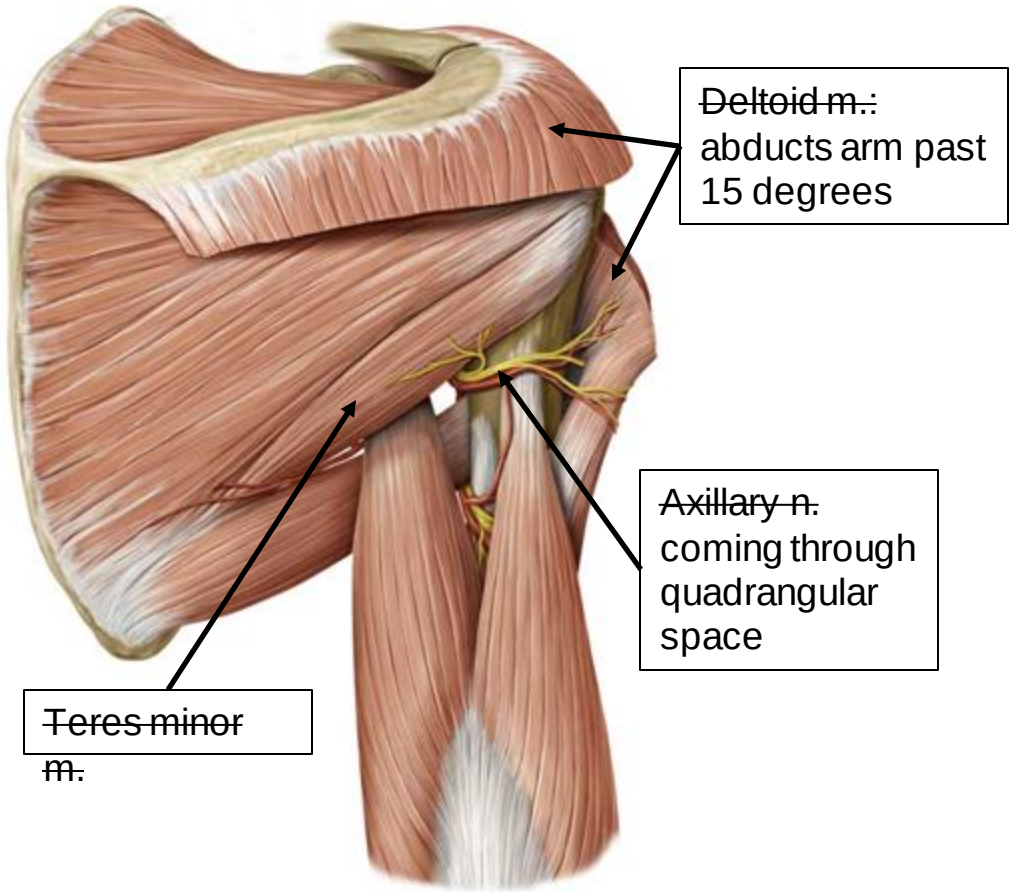
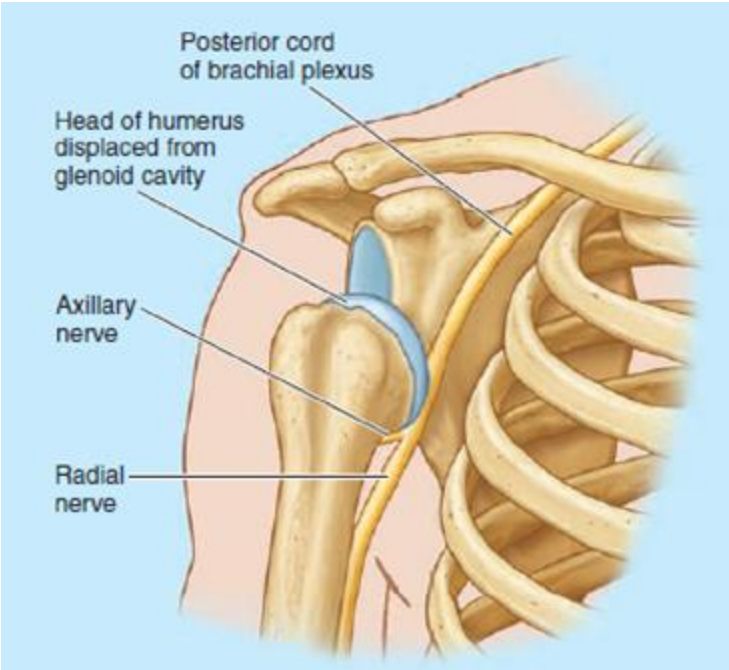
Medial border of scapula thrust posteriorly when patient pushes hand against wall



Damage to Axillary n. (C5-C6)

Occurs in quadrangular space from fracture of surgical neck of humerus or inferior dislocation of glenohumeral joint.

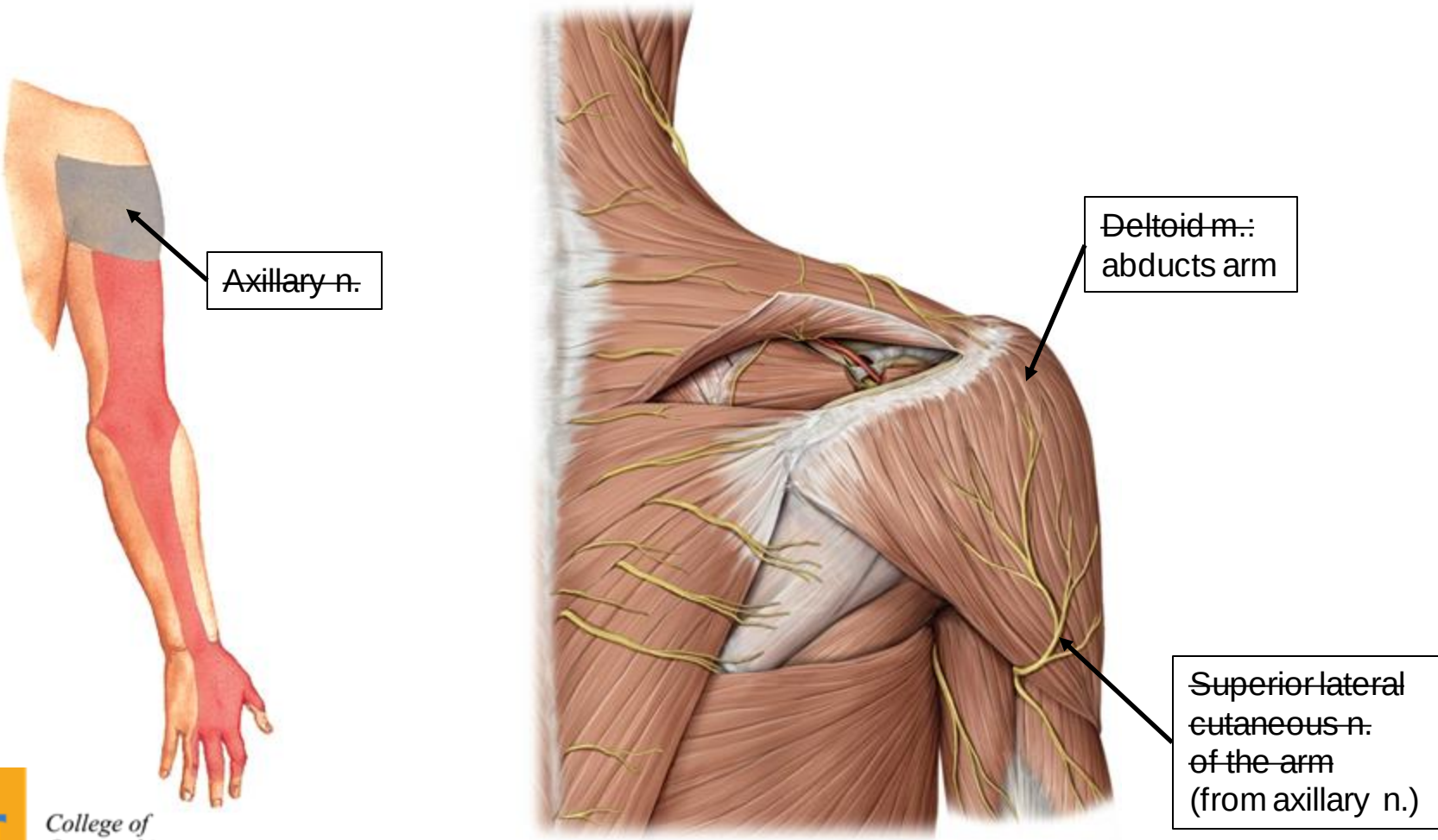
- 1. Loss of abduction of arm from deltoid m.
(loss of lateral rotation of arm from teres minor m., but compensated for by infraspinatus m.)
- 2. Loss of cutaneous sensation to proximal lateral arm



Damage to Axillary n. (C5-C6)

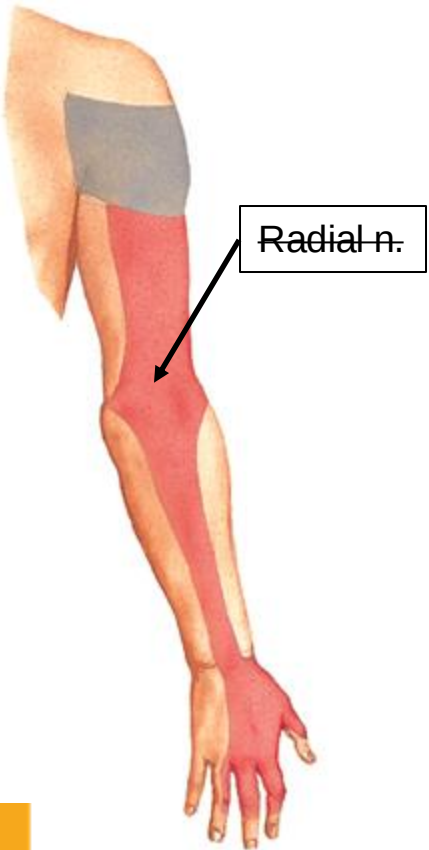
Occurs in quadrangular space from fracture of surgical neck of humerus or inferior dislocation of glenohumeral joint.

- 1. Severely reduced ability to abduct arm from deltoid m.
(loss of lateral rotation or arm from teres minor m., but compensated for by infraspinatus m.)
- 2. Loss of cutaneous sensation to upper lateral arm

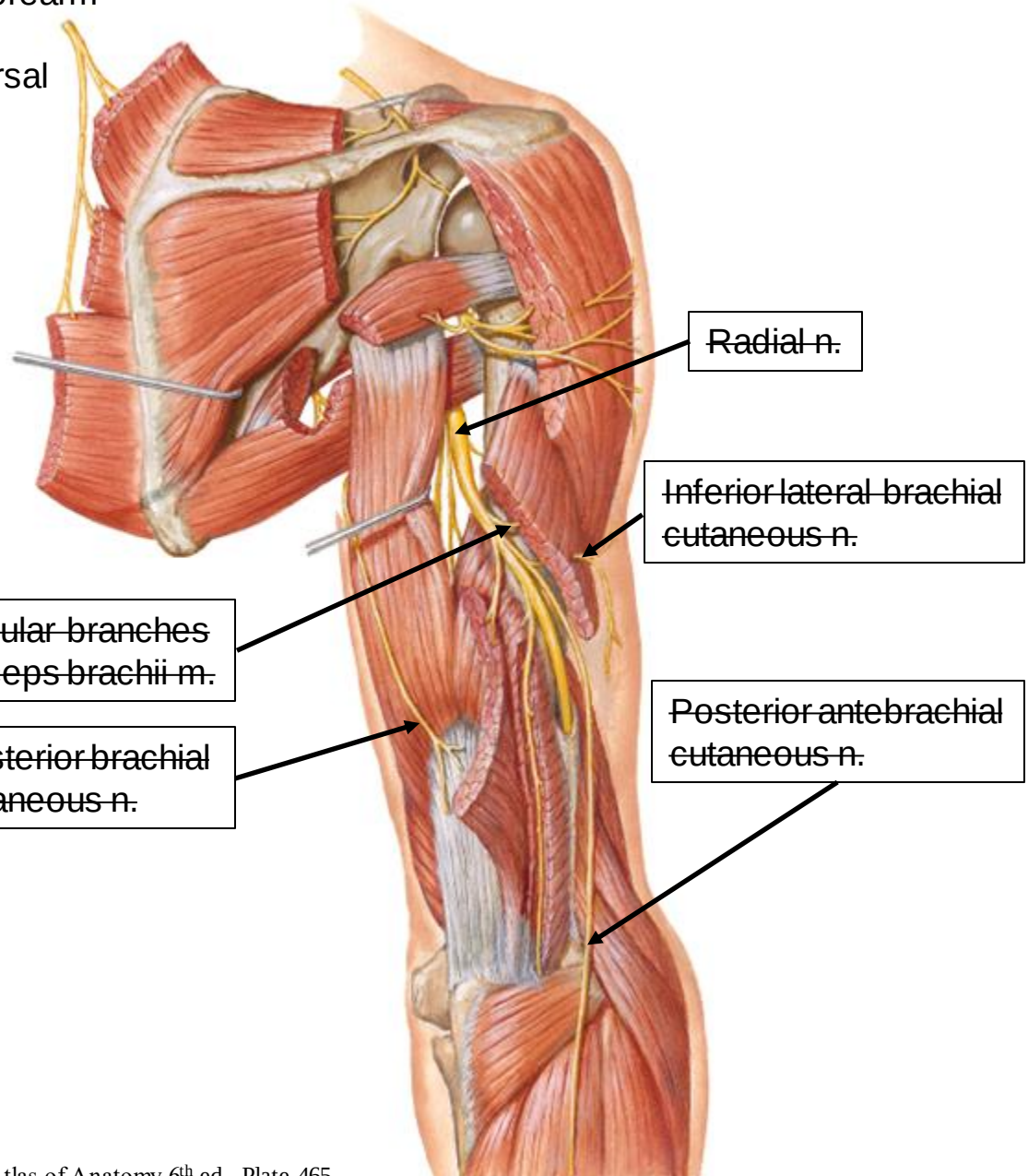


Damage to radial n. in axilla (C5-C8, T1):

Due to poorly fitted crutches or draping arm over back of chair (Saturday night palsy)
Innervation to extensors of posterior arm and forearm
Cutaneous innervation to posterior arm, and forearm, dorsolateral hand, proximal part of dorsal side of lateral 3.5 digits



Radial n.



Radial n.

Inferior lateral brachial cutaneous n.

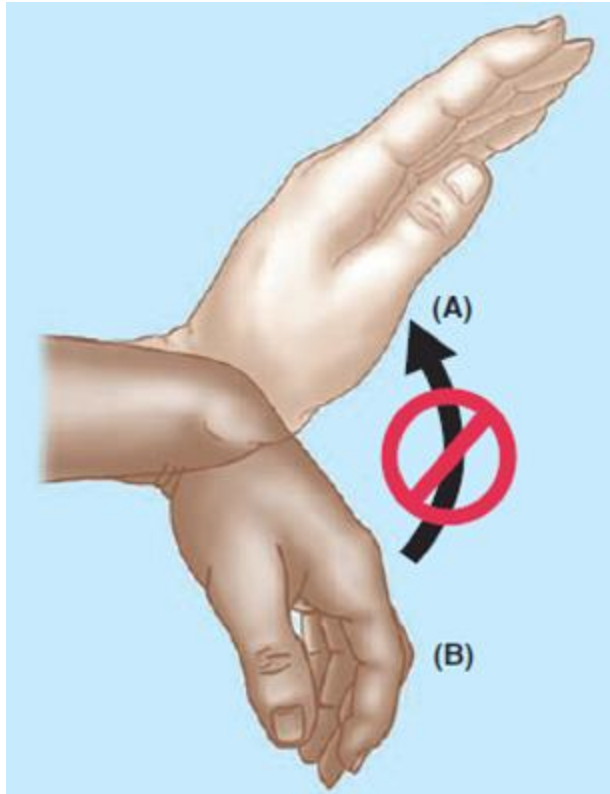
Muscular branches to triceps brachii m.

Posterior brachial cutaneous n.

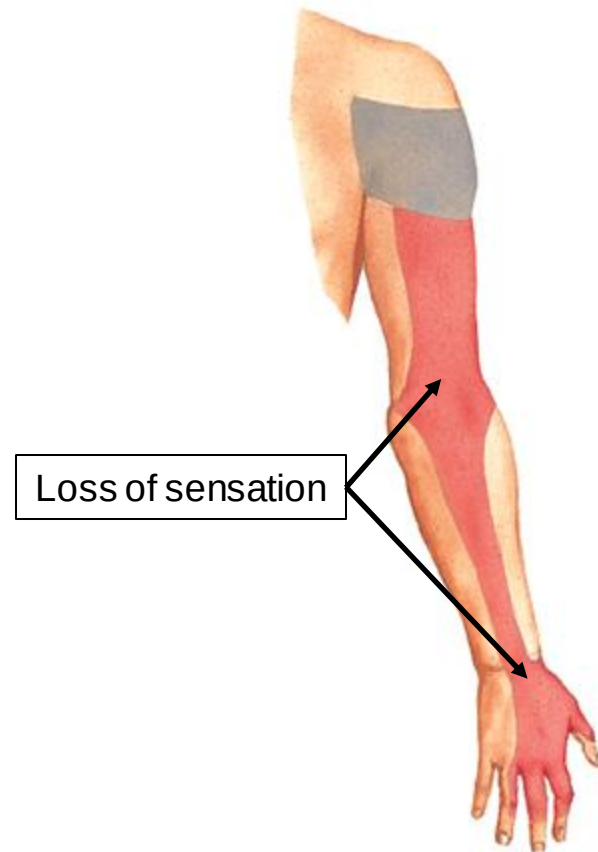
Posterior antebrachial cutaneous n.

Damage to radial n. in axilla (C5-C8, T1):

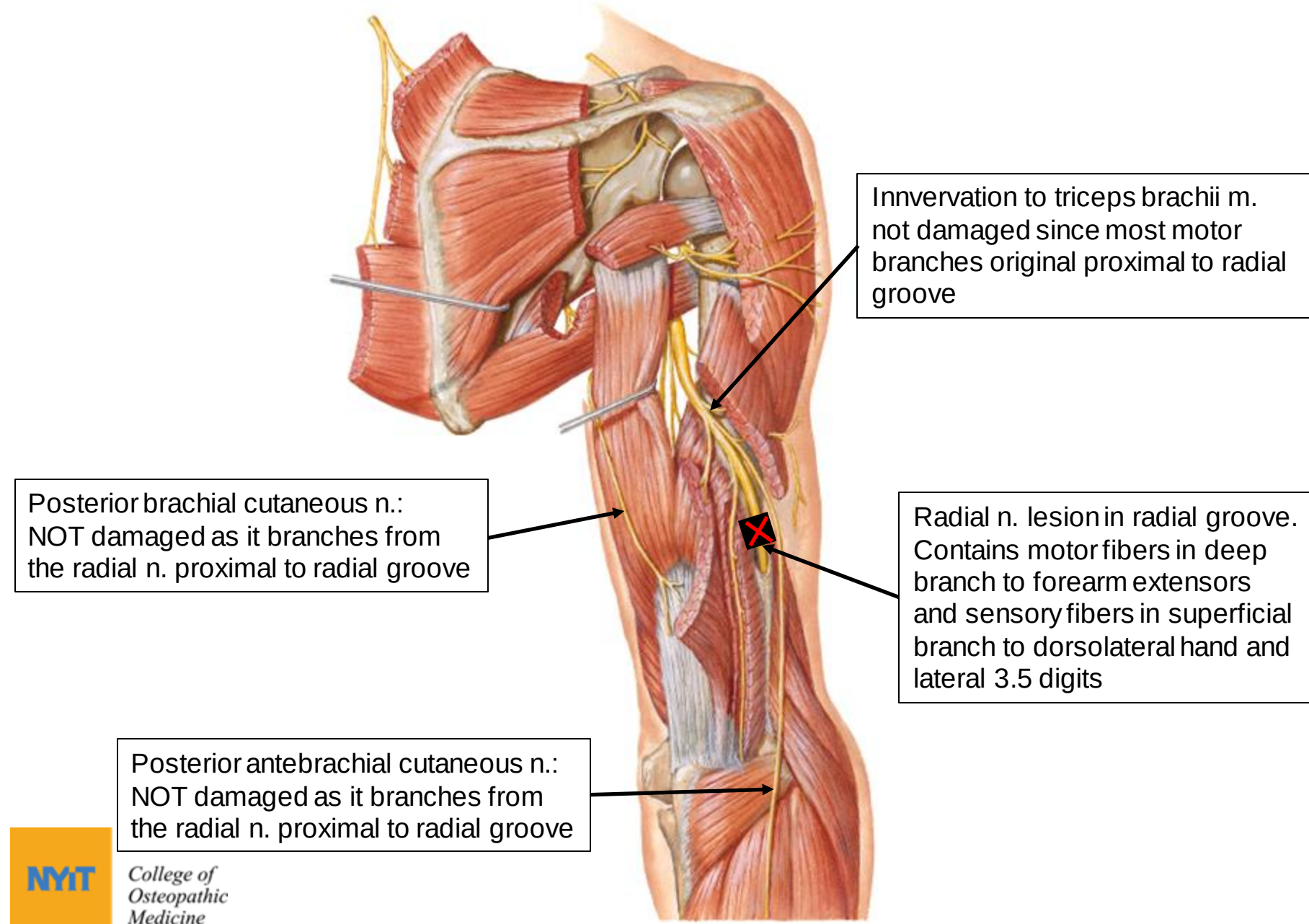
1. Loss of extension at elbow from inaction of triceps brachii m.
2. Loss of extension of wrist from inaction of extensors of hand (posterior forearm) results in unopposed action of long flexors causing wrist-drop
3. Loss of sensation from posterior arm, posterior forearm, dorsolateral hand, and dorsum of proximal part of lateral 3.5 digits



Wrist-drop:
Loss of extension in hand

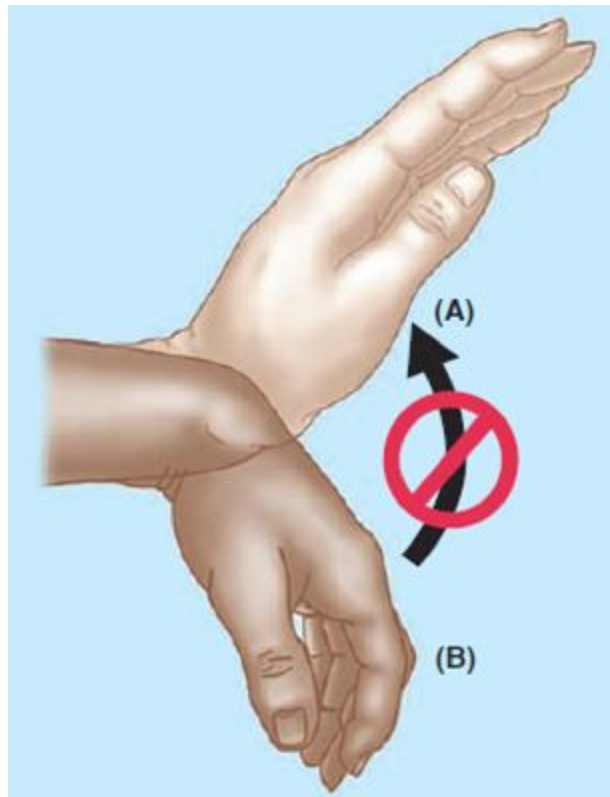


Damage to radial n. in radial groove due to fracture of mid-shaft of humerus

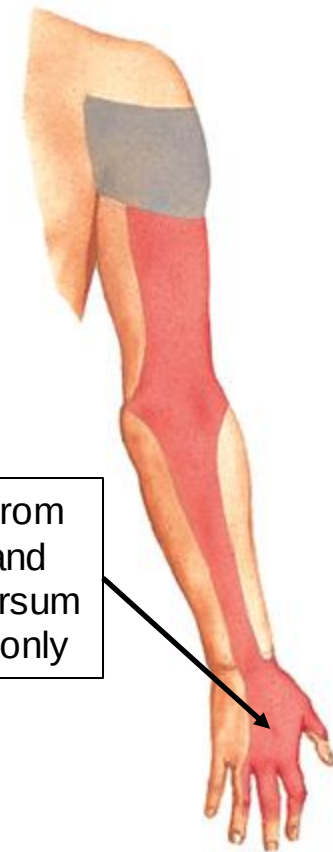


Damage to radial n. in radial groove (C5-C8, T1):

1. Loss of extension of wrist from inaction of extensors of hand (deep branch of radial n.) results in wrist-drop.
2. Triceps are not paralyzed as they are innervated in axilla (no loss of extension of elbow)
3. Loss of sensation from dorsolateral hand, and dorsum of proximal part of lateral 3.5 digits (superficial branch of radial n.)
4. Posterior arm and posterior forearm are unaffected as these sensory branches come off radial n. proximal to radial groove.

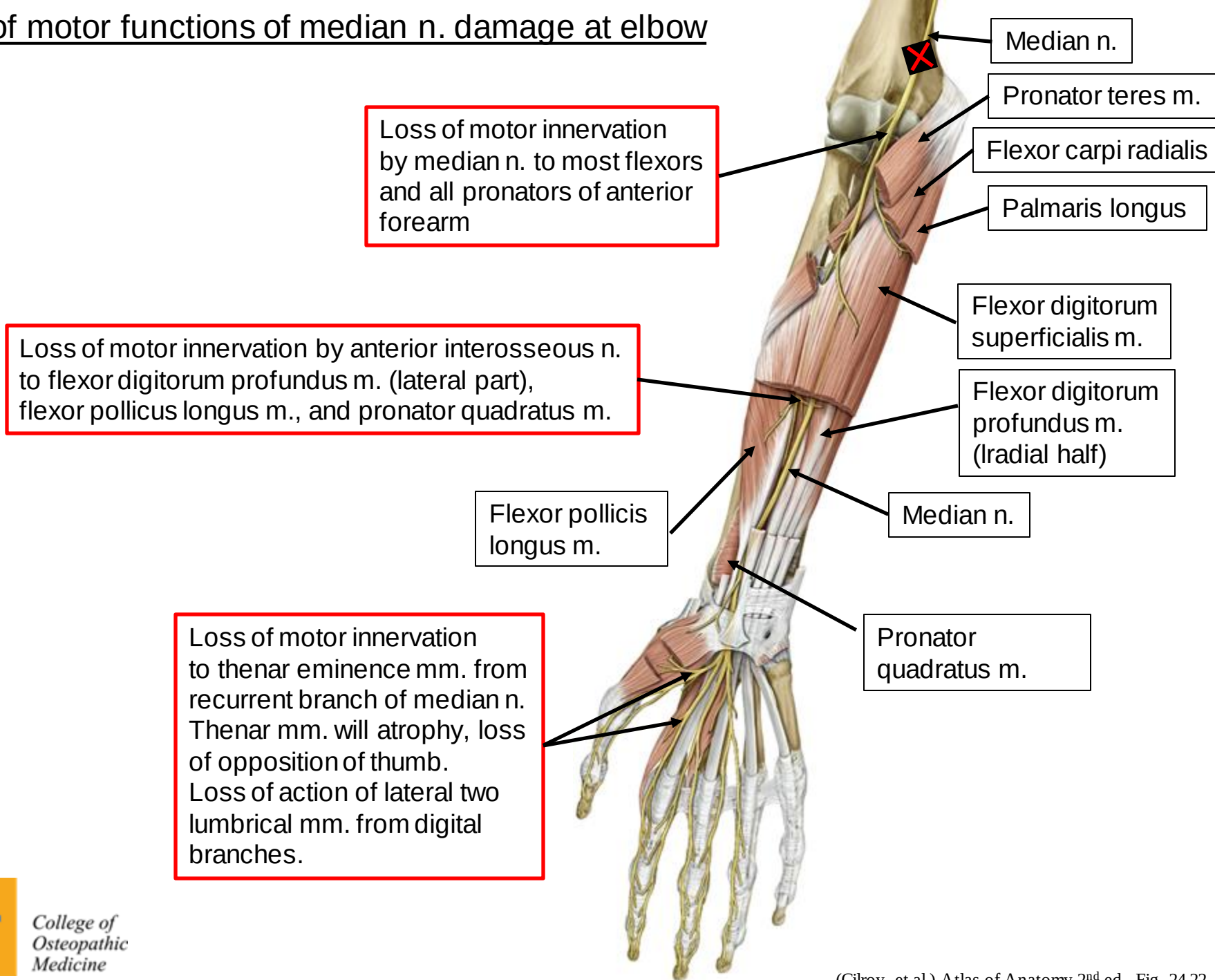


Wrist-drop:
Loss of extension in hand



Loss of sensation from
dorsolateral hand and
proximal part of dorsum
of lateral 3.5 digits only

Loss of motor functions of median n. damage at elbow



Result of damage to median n. at elbow when patient attempts to make a fist

Ulnar n. intact:
Flexion of and DIP joints
(and by extension PIP
joints) of digits 4 and 5
still possible due to intact
medial half of flexor
digitorum profundus m.

Loss of innervation to lateral two lumbricals,
lateral half of flexor digitorum profundus m.,
and all of flexor digitorum superficialis m.
prevents flexion of digits 2 and 3

Loss of innervation to
flexor pollicis longus m.
and flexor pollicis brevis
m. prevent flexion
of thumb.
Loss of opposition of
thumb.

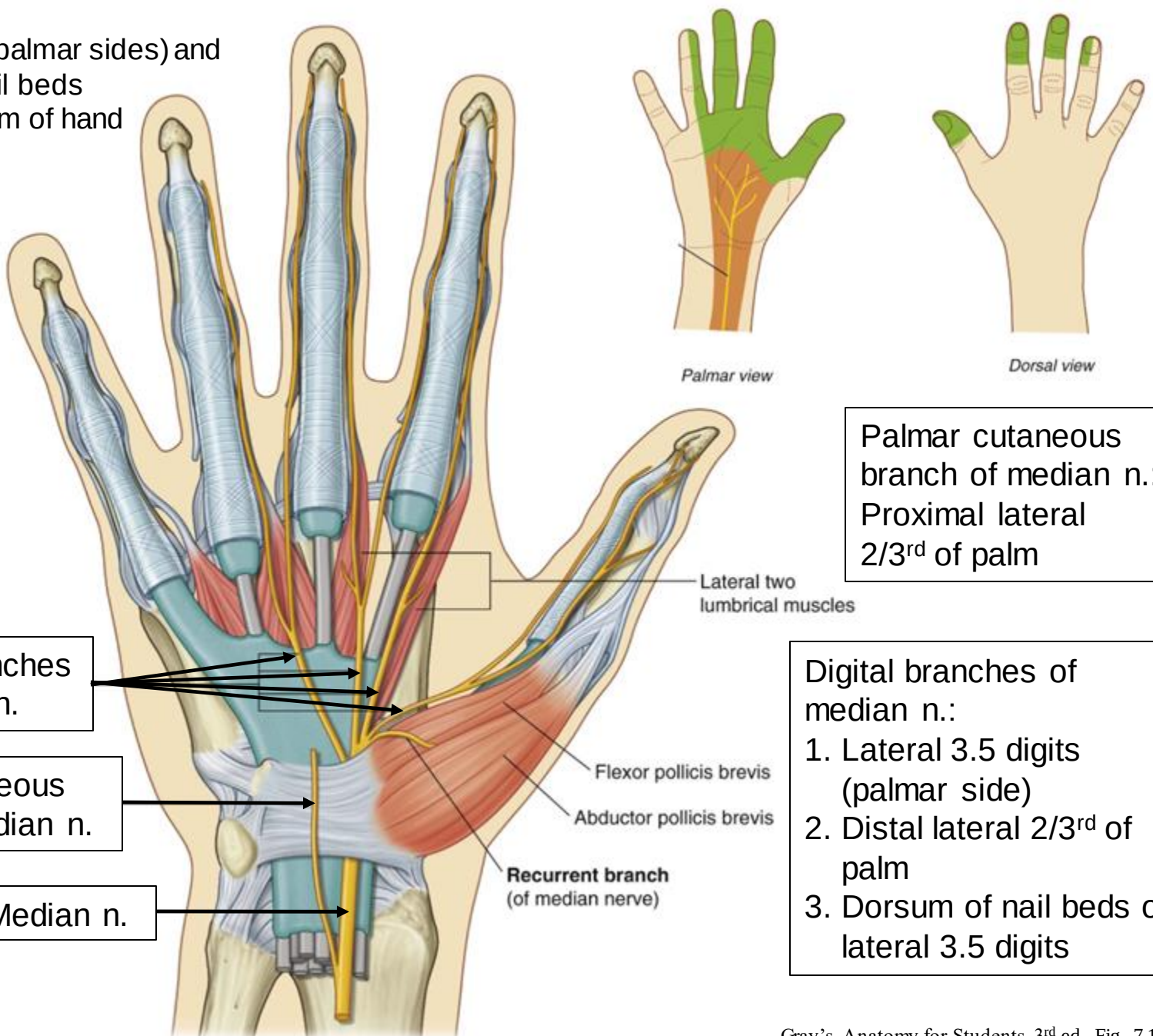


“Hand of benediction” Or ape hand
(a) Median nerve injury

Loss of sensory functions of median n. in hand from damage at elbow

Sensory innervation:

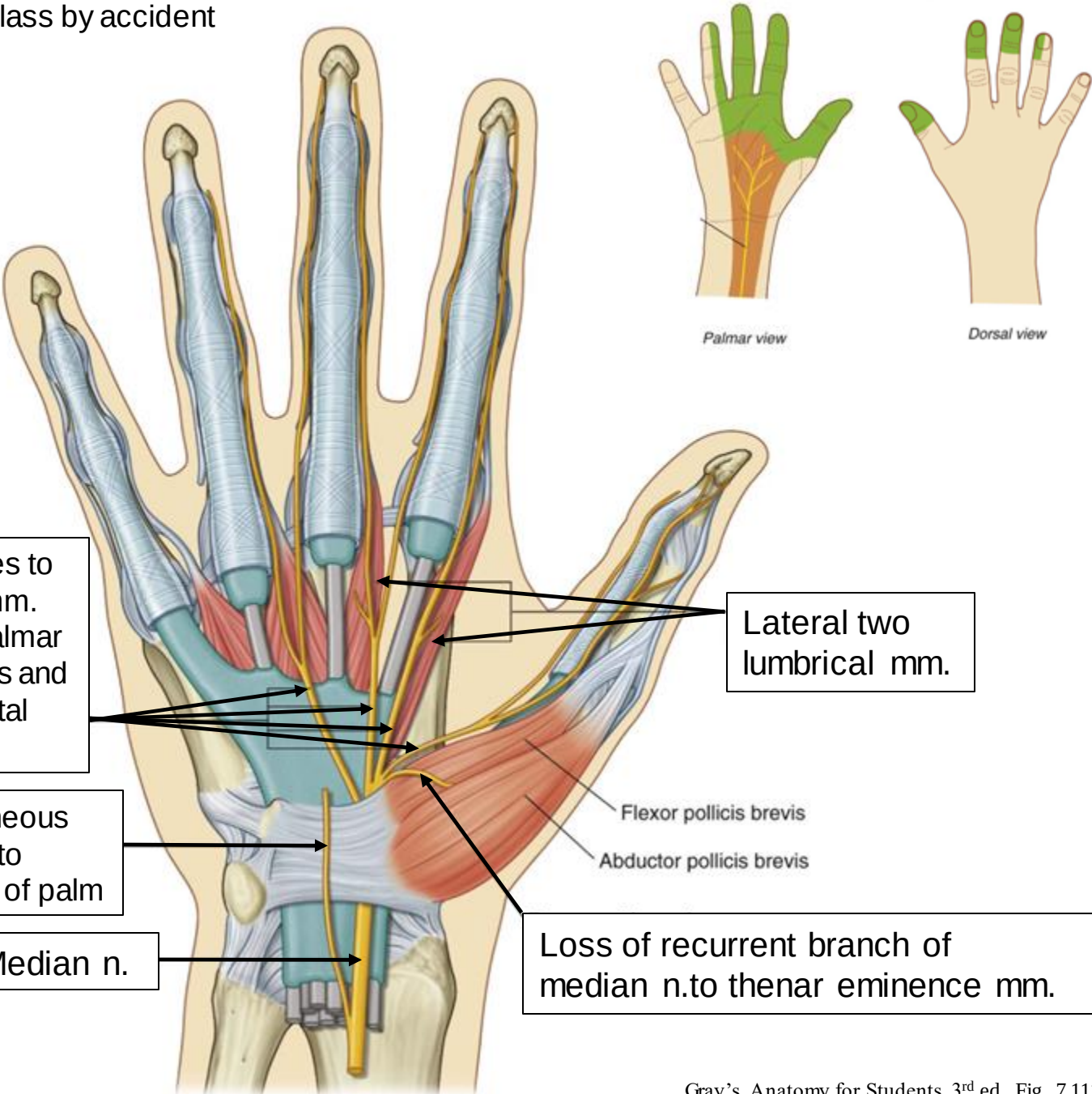
- 1. Lateral 3.5 digits (palmar sides) and dorsum of their nail beds
- 2. Lateral 2/3rd of palm of hand



- Loss of opposition of thumb
- Severe impairment of flexion of hand
 - Less impairment in digits 4-5 (ulnar n. intact)
 - Digits 2-3 have diminished ability to flex and extend
 - “Ape hand” “Hand of Benediction” presentation (be careful with these terms...they are used inconsistently)
- Loss of pronation
- Loss of sensation from lateral 2/3 of palm of hand, palmar side of lateral 3 ½ digits, and nail beds of lateral 3 ½ digits

Damage to median n. at wrist:

Laceration by knife or broken glass by accident or attempted suicide



Summary of consequences of damage to median n. at wrist:

Loss of motor innervation:

1. All three thenar eminence mm. lose innervation by recurrent branch of median n.
 - Loss of opposition and atrophy of thumb muscles (thenar eminence goes flat)
2. Lateral two lumbrical mm.
 - Much less significant deficits since interosseous mm. are innervated by ulnar n. (both flex MCP and extend PIP and DIP joints of digits 2 and 3).

Loss of sensory innervation same as damage to elbow:

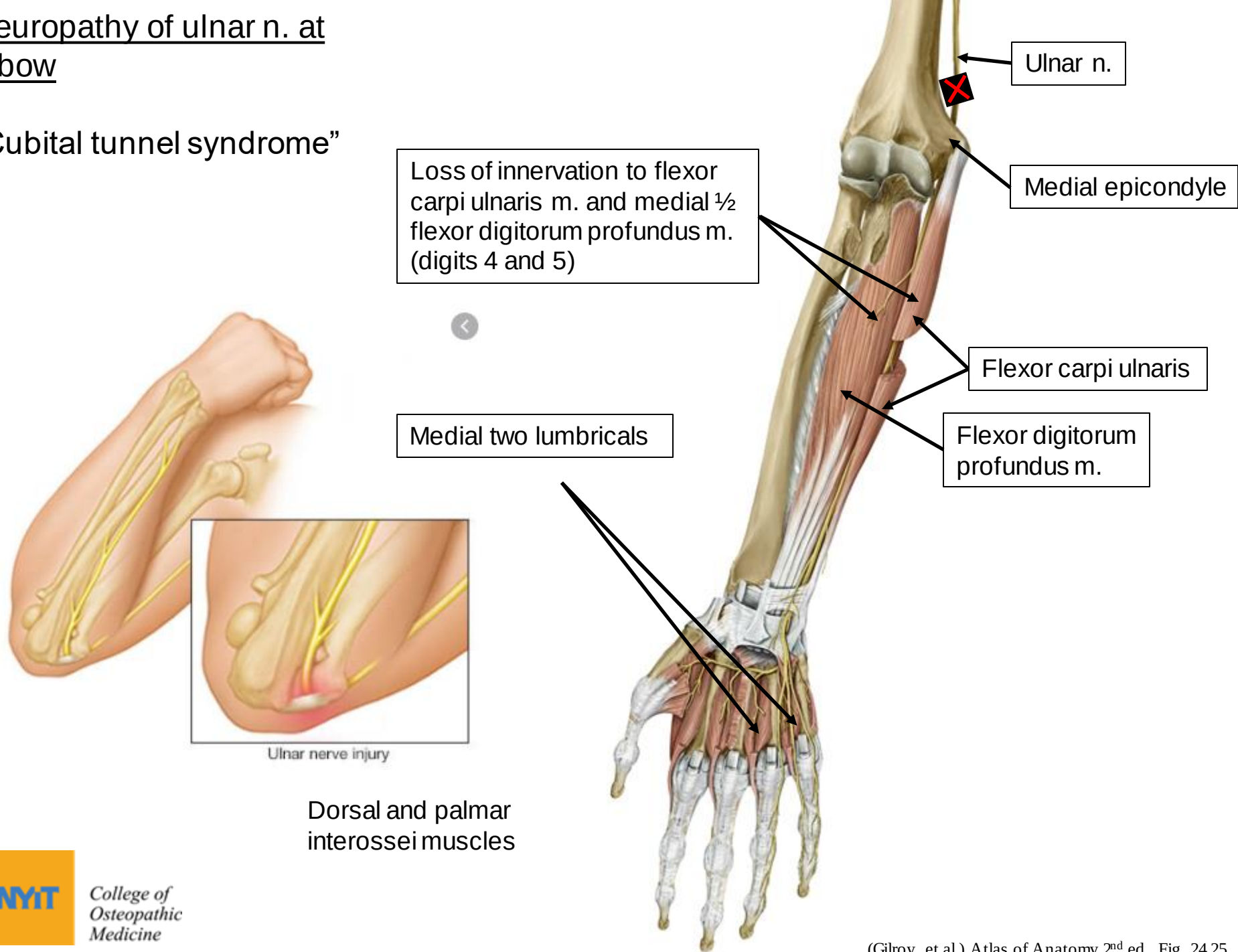
1. Lateral 3.5 digits (palmar side) and dorsum of their nail beds
2. Lateral 2/3rd of palm of hand



"Hand of benediction" Or ape hand
(a) Median nerve injury

Neuropathy of ulnar n. at elbow

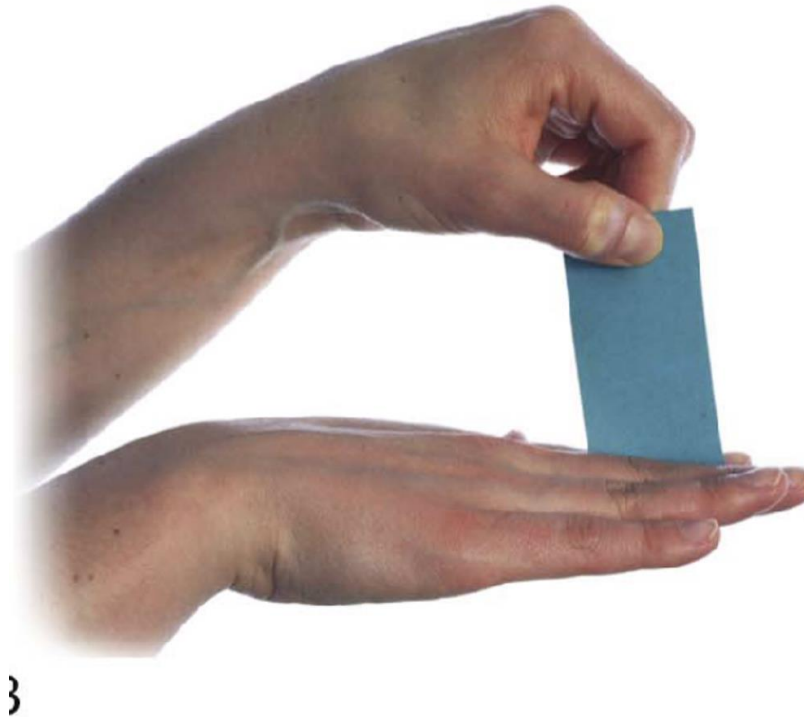
“Cubital tunnel syndrome”



Lesion of ulnar n. at elbow:

-Loss of action of all interosseous mm. and medial two lumbrical mm.

Loss of adduction-abduction
by interosseous mm:
Patient can not grip a piece of
paper between fingers

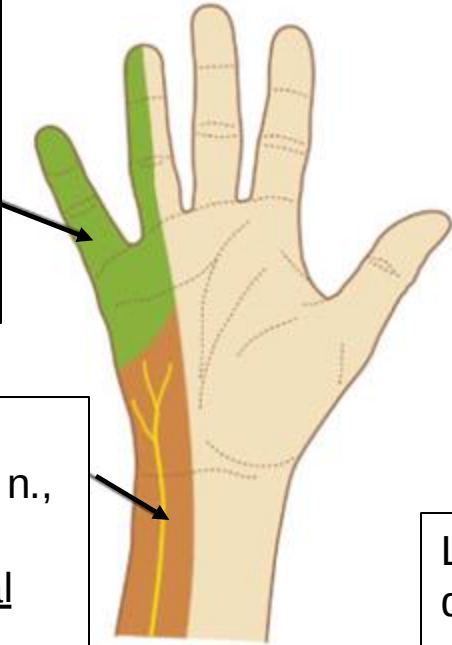


Lesion of ulnar n. at elbow:

Loss of cutaneous innervation to medial 1/3rd of hand and medial 1.5 fingers

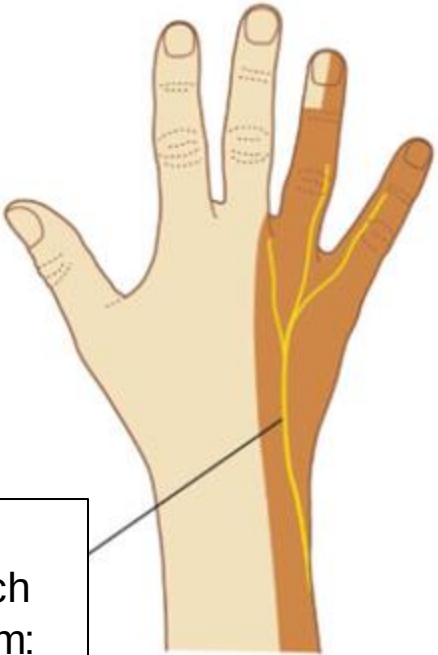
Loss of sensation from digital (superficial) branches of ulnar n.:
Distal 1/3rd of medial palm and medial 1.5 digits.

Loss of sensation from palmar branch of ulnar n., from forearm:
Proximal 1/3rd of medial palm



Palmar view

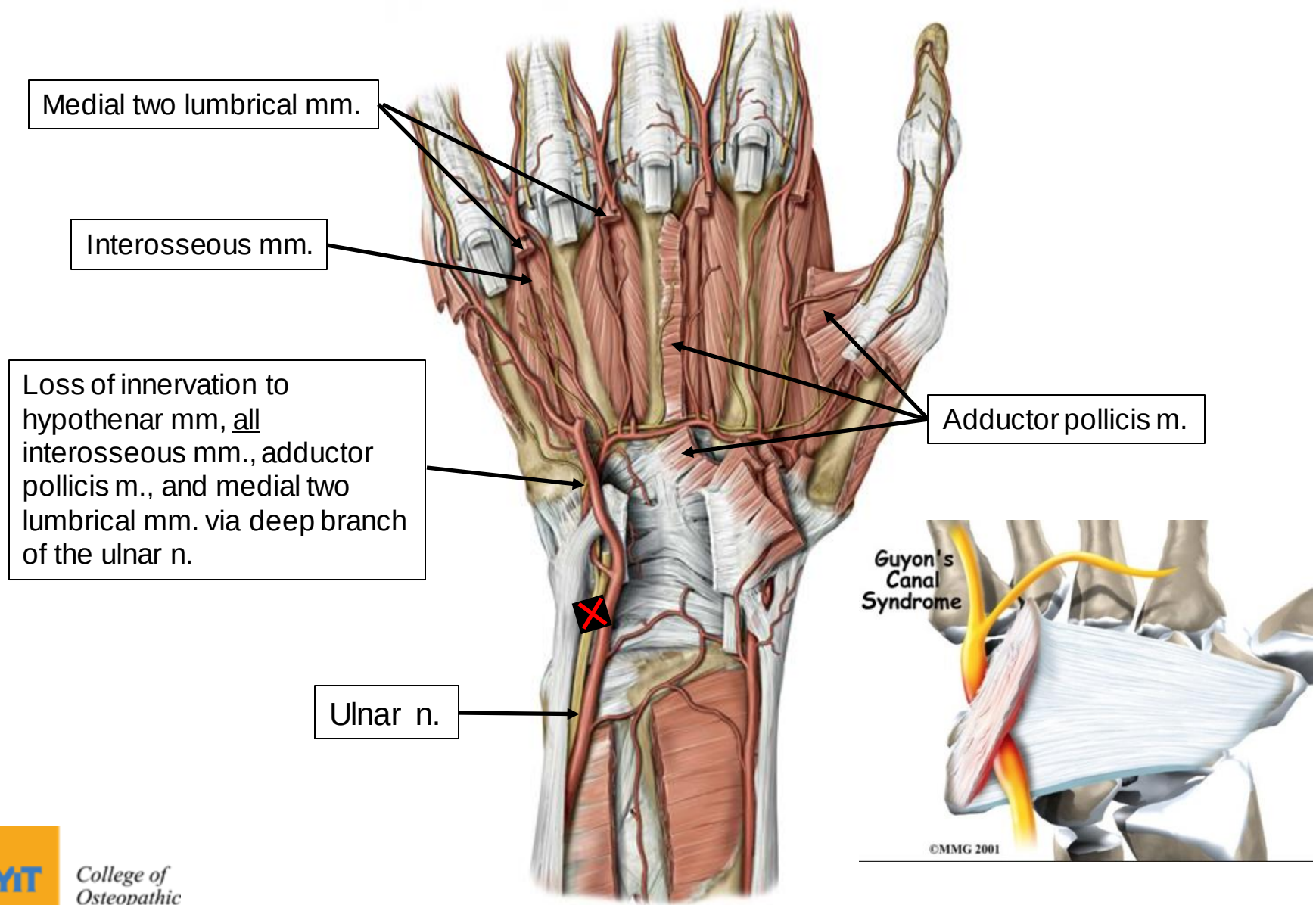
Loss of sensation from dorsal cutaneous branch of ulnar n., from forearm:
Medial 1/3rd of dorsum of hand and medial 1.5 digits



Dorsal view

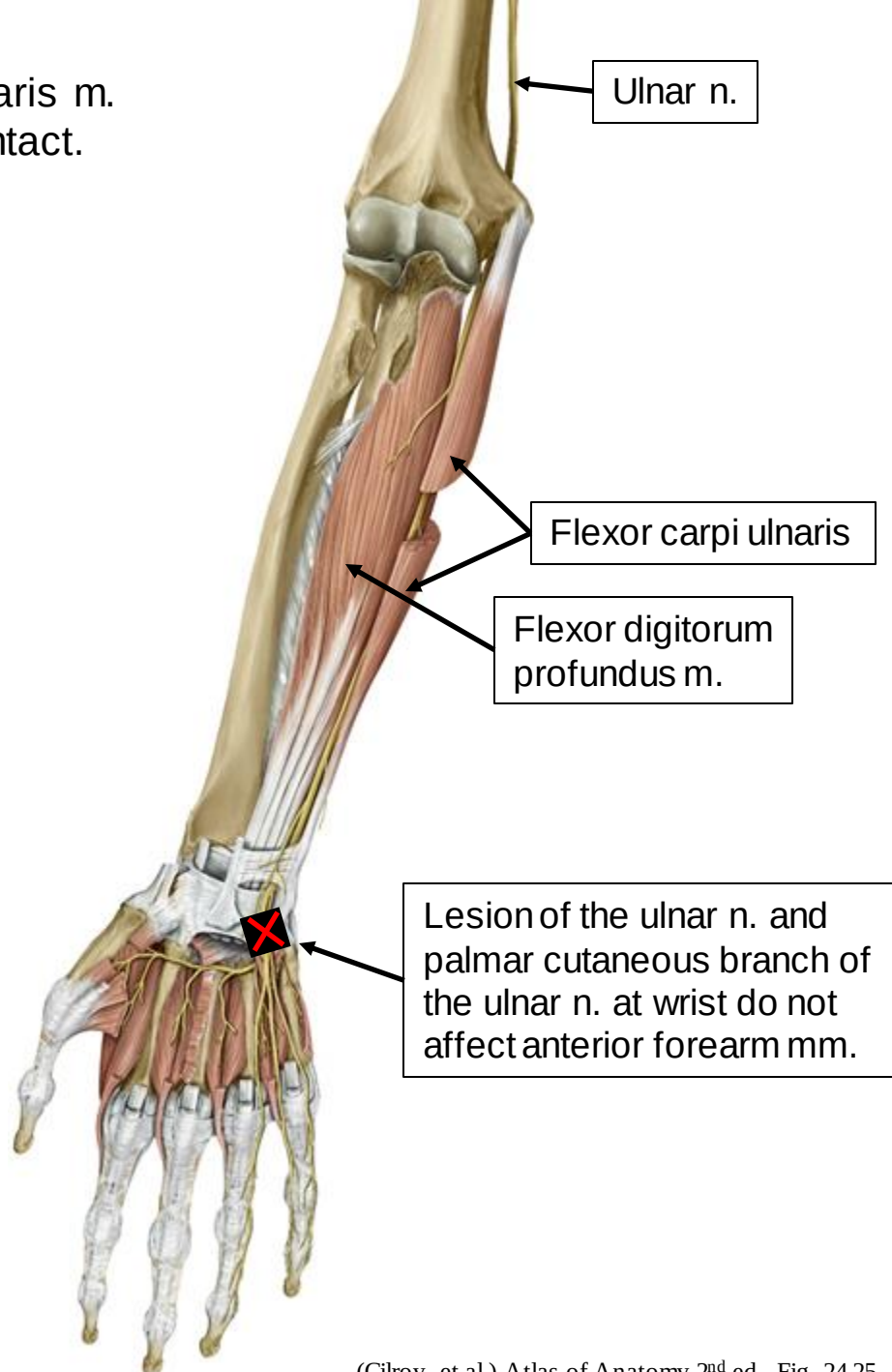
Neuropathy of ulnar n. at wrist:

- Guyon's Canal (ulnar tunnel) syndrome
- Motor loss in hand same as lesion at elbow



Neuropathy of ulnar n. at wrist:

Damage to ulnar n. at wrist leaves flexor carpi ulnaris m. and medial half of flexor digitorum profundus m. intact.
Wrist flexion NOT affected



Summary of motor damage to ulnar n. at wrist:

1. Paralysis of most intrinsic mm. of the hand
 - Exception-thenar eminence mm. and lateral two lumbrical mm.
 - Opposition of thumb intact
2. Consequences of paralysis:
 - Loss of adduction-abduction of fingers due to loss of interosseous mm.
 - Impairment of flexion of MCP joints and extension of PIP and DIP joints of digits 4-5 due to loss of interosseous mm. and medial two lumbrical mm.
3. Not affected:
 - Actions of flexor carpi ulnaris m. and all of flexor digitorum profundus m. intact due to ulnar n. innervation occurring above lesion at wrist.

Digits 2-3 not affected



Flexion of PIP and DIP joints of digits 4-5 due to loss of medial two lumbrical mm. and interosseous mm. and unopposed action of FDP and FDS

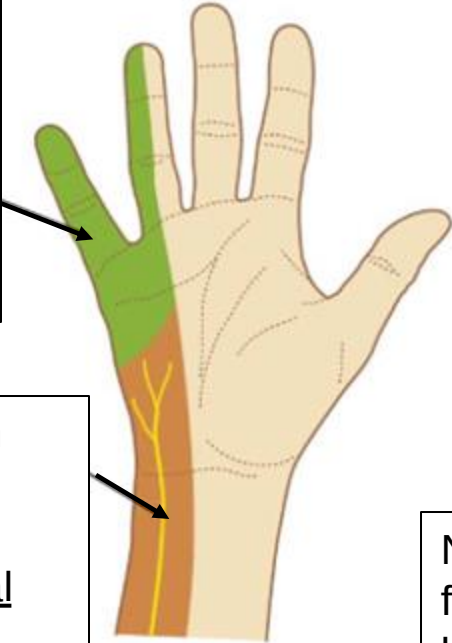
Hyperextension of MCP joints of digits 4-5 occurs due to unopposed action of extensor digitorum m. due to loss of interossei mm. and medial two lumbrical mm.

Lesion of ulnar n. at wrist:

Loss of cutaneous innervation to distal medial 1/3rd of hand and medial 1.5 fingers

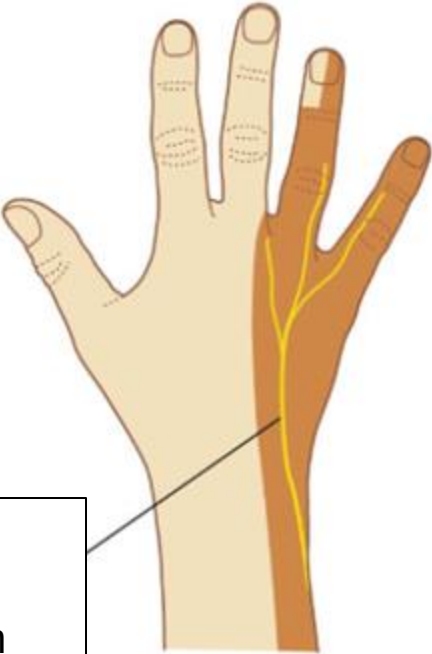
Loss of sensation from digital (superficial) branches of ulnar n.:
Distal 1/3rd of medial palm and medial 1.5 digits.

NO LOSS of sensation from palmar branch of ulnar n., from forearm:
Proximal 1/3rd of medial palm



Palmar view

NO LOSS of sensation from dorsal cutaneous branch of ulnar n., from forearm:
Medial 1/3rd of dorsum of hand and medial 1.5 digits



Dorsal view

Summary of different claw effects from ulnar n. lesions at different levels

1. Klumpke's Palsy (damage to inferior trunk of brachial plexus)

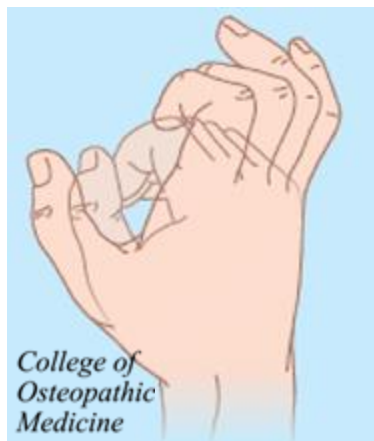
- All four MCP joints extended due to loss of all intrinsic muscles of the hand from C8-T1 fibers from median and ulnar nn. and unopposed action of extensor digitorum m.
- PIP joints of digits 2-3 flexed due to unopposed action of flexor digitorum superficialis, DIP joint 2-3 flexion depends on functionality of lateral half of flexor digitorum profundus and severity of injury.
- PIP of digits 4-5 flexed because of unopposed action of FDS, DIP not flexed because no action from medial side of FDP.
- Result: total claw hand, digits 2-3 possibly clawed worse.

2. High lesion of ulnar n. (i.e. at elbow region, broken humerus or cubital tunnel syndrome)

- MCP of digits 4-5 extended because of loss of interosseous mm. and medial two lumbrical mm.
- MCP of digits 2-3 not extended because lateral two lumbrical mm. are still intact.
- PIP and DIP of digits 2-3 not flexed because lateral lumbrical mm. are still intact.
- PIP and DIP of digits 4-5 same as above. PIP flexed, DIP not flexed.
- Result: "ulnar paradox", higher lesion (=more serious) than wrist lesion, but claw effect is lessened due to lack of action from lateral side of FDP. Claw effect will worsen as nerve heals and action of FDP is restored.

3. Low lesion of ulnar n. (i.e. at wrist, Guyon's canal (ulnar tunnel) entrapment)

- Same as above, except both PIP and DIP of digits 4-5 are flexed because of unopposed action of FDP.
- Result: Ulnar claw





Brachial Plexus: Lesions and Neuropathies

Principles and Practices of Osteopathic Medicine I

2020

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Part 5: example cases, questions and summary

A 21-year-old male varsity collegiate wrestler reported left elbow pain and loss of strength and motor control in his left upper limb, most notably in his wrist and hand. He also reports tingling sensations on digit 5 and the medial half of digit 4.

He did not recall experiencing a specific injury that caused the problem. However, he complains that the symptoms worsen after wrestling practice. Point tenderness was found along the medial midshaft of the humerus, the medial epicondyle, and the olecranon process.

Q1.

Which injury is most likely?

- a. cubital tunnel syndrome
- b. Erb-Duchenne palsy
- c. radial neuropathy within the axilla
- d. carpal tunnel syndrome
- e. axillary nerve lesion

Q2.

Which motor function do you expect the patient to mostly likely have difficulty with?

- a. extension of digits 1 and 2
- b. extension of the wrist
- c. abduction of the arm
- d. pronation
- e. adduction of the fingers

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Which injury is most likely?

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- b. Erb-Duchenne palsy
- c. radial neuropathy within the axilla
- d. carpal tunnel syndrome
- e. axillary nerve lesion in the quadrangular space

Q2.

Which motor function do you expect the patient to have difficulty with?

- a. extension of digits 1 and 2
- b. extension of the wrist
- c. abduction of the arm
- d. pronation
- e. *adduction of the fingers

A 25 day-old neonate is presented to the ED. The parents are concerned about the lack of movement in the right upper limb.

A few days after birth, the patient underwent a right postero-lateral thoracotomy to repair an esophageal birth defect. At presentation, the neonate's upper right limb maintains an adducted and pronated posture. Shoulder abduction, supination, and flexion of the elbow were not observed; however, the patient is capable of palmar grasp and distal finger movements.



Q1

A lesion of which nerve best explains this presentation?

- a. inferior trunk of brachial plexus
- b. median nerve
- c. long thoracic nerve
- d. superior trunk of brachial plexus
- e. ulnar nerve

Q2

Based on the neonate's presentation, where might you expect to find a sensory deficiency?

- a. medial side of forearm and hand
- b. lateral side of shoulder
- c. lateral side of arm, forearm and hand
- d. 5th digit and medial half of 4th digit
- e. axilla

A 25 day-old neonate is presented to the ED. The parents are concerned about the lack of movement in the right upper limb.

A few days after birth, the patient underwent a right postero-lateral thoracotomy to repair an esophageal birth defect. At presentation, the neonate's upper right limb maintains an adducted and pronated posture. Shoulder abduction, supination, and flexion of the elbow were not observed; however, the patient is capable of palmar grasp and distal finger movements.



Q1

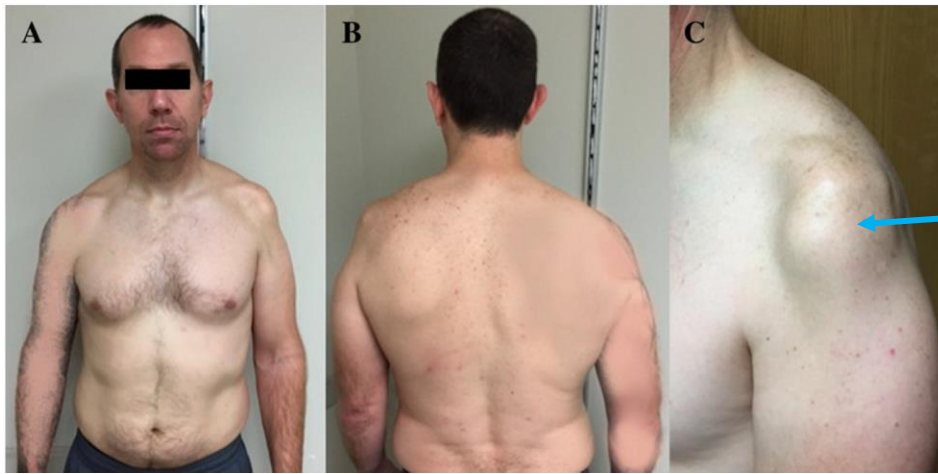
A lesion of which nerve best explains this presentation?

- a. inferior trunk of brachial plexus
- b. median nerve
- c. long thoracic nerve
- d. *superior trunk of brachial plexus
- e. ulnar nerve

Q2

Based on the neonate's presentation, where might you expect to find a sensory deficiency?

- a. medial side of forearm and hand
- b. lateral side of shoulder
- c. *lateral side of arm, forearm and hand
- d. 5th digit and medial half of 4th digit
- e. axilla



Left deltoid muscle atrophy

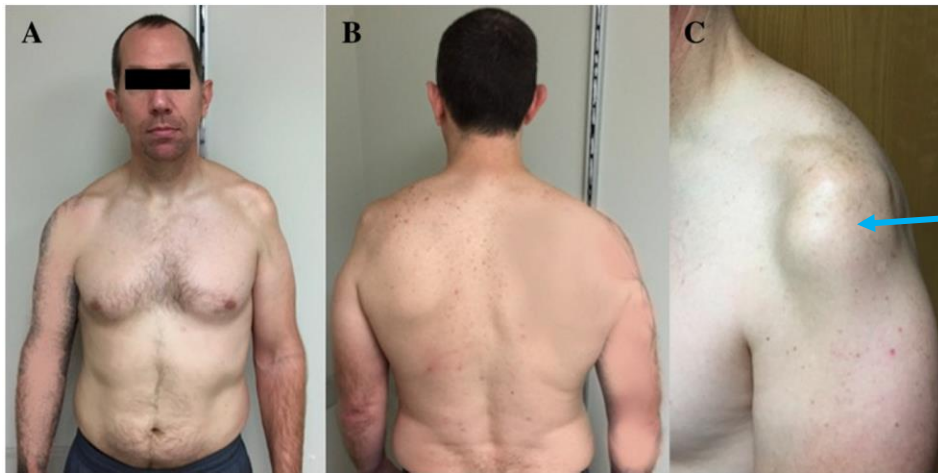
A 43 year-old man presents with conspicuous atrophy of his left deltoid muscle. MRI reveals atrophy of both the deltoid and teres minor. No other notable muscular atrophy was found. He says his deformity is a result of shoulder injury sustained in a car accident 25 years ago. Although he reports some upper limb weakness, he has learned to compensate for his injury and is able to function without any major disabilities.

Q1. Which nerve was injured in the car accident?

- a. long thoracic
- b. superior trunk
- c. median
- d. axillary
- e. suprascapular

Q2. What muscle enables the patient to abduct his left arm?

- a. triceps
- b. supraspinatus
- c. subscapularis
- d. coracobrachialis
- e. rhomboid major



Left deltoid muscle atrophy

A 43 year-old man presents with conspicuous muscular atrophy on his left shoulder. MRI reveals atrophy of both the deltoid and teres minor. No other notable muscular atrophy was discovered. He has diminished cutaneous sensation on the lateral side of his left shoulder. He says his deformity is a result of shoulder injury sustained in a car accident 25 years ago. Although he reports some upper limb weakness, he has learned to compensate for his injury and feels that he is able to function without any major disability.

Q1. Which nerve was injured in the car accident?

- a. long thoracic
- b. superior trunk
- c. median
- d. *axillary
- e. suprascapular

Q2. What muscle enables the patient to abduct his left arm?

- a. triceps
- b. *supraspinatus
- c. subscapularis
- d. coracobrachialis
- e. rhomboid major

Nerve Injuries of the Upper Limb

- Injury to spinal nerves: dermatomes and myotomes
- Injury to proximal brachial plexus
 - Superior trunk
 - Inferior trunk
- Injury to the long thoracic nerve (nerve to serratus anterior)
- Injury to the axillary nerve
- Injury to the radial nerve
 - Different effects depending on location of injury
- Injury to median nerve
 - Different effects depending on location of injury
- Injury to ulnar nerve
 - Different effects depending on location of injury